

DATA ANALYTICS WITH COGNOS - GROUP 5

PROJECT: CUSTOMER CHURN PREDICTION

PHASE 5 : PROJECT DOCUMENTATION AND SUBMISSION

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PHASE 5 : FINAL DOCUMENT SUBMISSION



INTRODUCTION:

In today's highly competitive business landscape, retaining customers is paramount for the sustained success of any organization. Customer churn, the phenomenon of customers discontinuing their association with a company, poses a significant challenge to businesses across various industries. The ability to predict and mitigate customer churn is crucial for maintaining revenue, sustaining growth, and ensuring customer satisfaction.

This project is dedicated to the task of Customer Churn Prediction, a data-driven approach that harnesses the power of advanced analytics and machine learning to foresee which customers are at risk of churning. By proactively identifying customers who are likely to leave, businesses can take timely measures to retain them, thus reducing revenue loss and improving customer retention rates.

In this endeavor, we will leverage historical customer data, employ cutting-edge machine learning algorithms, and conduct in-depth analysis to build a predictive model. The objective is to not only identify potential churners but also to understand the underlying factors contributing to churn. This insight can inform targeted retention strategies, product enhancements, and customer relationship management, ultimately improving customer loyalty and profitability.

Through this project, we aim to empower businesses to be more customer-centric, data-savvy, and proactive in managing their customer relationships. The journey begins with data exploration, feature engineering, model development, and culminates in a robust churn prediction system that can drive informed decision-making and foster long-term customer satisfaction.

OBJECTIVES

The primary objective for a customer churn prediction project is to develop a predictive model that can accurately identify customers who are likely to churn or discontinue their relationship with a business.

- Predict whether customers will continue to use the service or will leave the service.
- Understanding the customer behaviors:

- What keeps customers using the service
- What makes customers leave the service

DATA SOURCE

Telco customer churn:

The dataset contains 7043 rows and 21 features. Each row represents a customer, each column contains the customer's attributes described in the column Metadata

Dataset link : <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	customerID	gender	SeniorCitiz	Partner	Dependents	tenure	PhoneServ	MultipleLines	InternetSer	OnlineSecurity	OnlineBackup	DeviceProte	TechSupp	Streaming	Streaming	Contract	PaperlessP	PaymentMeth	MonthlyCharge	TotalCharge	Churn
2	7590-VHVI	Female	0	Yes	No	1	No	No phone sen	DSL	No	Yes	No	No	No	No	Month-to-+	Yes	Electronic che	29.85	29.85	No
3	5575-GNVI	Male	0	No	No	34	Yes	No	DSL	Yes	No	Yes	No	No	No	One year	No	Mailed check	56.95	1889.5	No
4	3668-QPYI	Male	0	No	No	2	Yes	No	DSL	Yes	Yes	No	No	No	No	Month-to-+	Yes	Mailed check	53.85	108.15	Yes
5	7795-CFOI	Male	0	No	No	45	No	No phone sen	DSL	Yes	No	Yes	Yes	No	No	One year	No	Bank transfer	42.3	1840.75	No
6	9237-HQIT	Female	0	No	No	2	Yes	No	Fiber optic	No	No	No	No	No	No	Month-to-+	Yes	Electronic che	70.7	151.65	Yes
7	9305-CDSH	Female	0	No	No	8	Yes	Yes	Fiber optic	No	No	Yes	No	Yes	Yes	Month-to-+	Yes	Electronic che	99.65	820.5	Yes
8	1452-KIOV	Male	0	No	Yes	22	Yes	Yes	Fiber optic	No	Yes	No	No	Yes	No	Month-to-+	Yes	Credit card (ai	89.1	1949.4	No
9	6713-OKO	Female	0	No	No	10	No	No phone sen	DSL	Yes	No	No	No	No	No	Month-to-+	No	Mailed check	29.75	301.9	No
10	7892-POO	Female	0	Yes	No	28	Yes	Yes	Fiber optic	No	No	Yes	Yes	Yes	Yes	Month-to-+	Yes	Electronic che	104.8	3046.05	Yes
11	6388-TABK	Male	0	No	Yes	62	Yes	No	DSL	Yes	Yes	No	No	No	No	One year	No	Bank transfer	56.15	3487.95	No
12	9763-GRSM	Male	0	Yes	Yes	13	Yes	No	DSL	Yes	No	No	No	No	No	Month-to-+	Yes	Mailed check	49.95	587.45	No
13	7469-LKBC	Male	0	No	No	16	Yes	No	No	No internet ser	No internet se	No internet	No interne	No interne	No interne	Two year	No	Credit card (ai	18.95	326.8	No
14	8091-TTVF	Male	0	Yes	No	58	Yes	Yes	Fiber optic	No	No	Yes	No	Yes	Yes	One year	No	Credit card (ai	100.35	5681.1	No
15	0280-XJGE	Male	0	No	No	49	Yes	Yes	Fiber optic	No	Yes	Yes	No	Yes	Yes	Month-to-+	Yes	Bank transfer	103.7	5036.3	Yes
16	5129-JLPI	Male	0	No	No	25	Yes	No	Fiber optic	Yes	No	Yes	Yes	Yes	Yes	Month-to-+	Yes	Electronic che	105.5	2686.05	No
17	3655-SNQI	Female	0	Yes	Yes	69	Yes	Yes	Fiber optic	Yes	Yes	Yes	Yes	Yes	Yes	Two year	No	Credit card (ai	113.25	7895.15	No
18	8191-XWSI	Female	0	No	No	52	Yes	No	No	No internet ser	No internet se	No internet	No interne	No interne	No interne	One year	No	Mailed check	20.65	1022.95	No
19	9959-WOFI	Male	0	No	Yes	71	Yes	Yes	Fiber optic	Yes	No	Yes	No	Yes	Yes	Two year	No	Bank transfer	106.7	7382.25	No
20	4190-MFLI	Female	0	Yes	Yes	10	Yes	No	DSL	No	No	Yes	Yes	No	No	Month-to-+	No	Credit card (ai	55.2	528.35	Yes
21	4183-MYFI	Female	0	No	No	21	Yes	No	Fiber optic	No	Yes	Yes	No	No	Yes	Month-to-+	Yes	Electronic che	90.05	1862.9	No
22	8779-QRD	Male	1	No	No	1	No	No phone sen	DSL	No	No	Yes	No	No	Yes	Month-to-+	Yes	Electronic che	39.65	39.65	Yes
23	1680-VDCI	Male	0	Yes	No	12	Yes	No	No	No internet ser	No internet se	No internet	No interne	No interne	No interne	One year	No	Bank transfer	19.8	202.25	No
24	1066-JKSG	Male	0	No	No	1	Yes	No	No	No internet ser	No internet se	No internet	No interne	No interne	No interne	Month-to-+	No	Mailed check	20.15	20.15	Yes
25	3638-WEA	Female	0	Yes	No	58	Yes	Yes	DSL	No	Yes	No	Yes	No	No	Two year	Yes	Credit card (ai	59.9	3505.1	No
26	6322-HRPI	Male	0	Yes	Yes	49	Yes	No	DSL	Yes	Yes	No	Yes	No	No	Month-to-+	No	Credit card (ai	59.6	2970.3	No
27	6865-JZNK	Female	0	No	No	30	Yes	No	DSL	Yes	Yes	No	No	No	No	Month-to-+	Yes	Bank transfer	55.3	1530.6	No
28	6467-CHFZ	Male	0	Yes	Yes	47	Yes	Yes	Fiber optic	No	Yes	No	No	Yes	Yes	Month-to-+	Yes	Electronic che	99.35	4749.15	Yes
29	8665-UTDI	Male	0	Yes	Yes	1	No	No phone sen	DSL	No	Yes	No	No	No	No	Month-to-+	No	Electronic che	30.2	30.2	Yes
30	5248-YGUI	Male	0	Yes	No	72	Yes	Yes	DSL	Yes	Yes	Yes	Yes	Yes	Yes	Two year	Yes	Credit card (ai	90.25	6369.45	No
31	8773-HHU	Female	0	No	Yes	17	Yes	No	DSL	No	No	No	No	Yes	Yes	Month-to-+	Yes	Mailed check	64.7	1093.1	Yes

The project involves using IBM Cognos to predict customer churn and identify factors influencing customer retention. The goal is to help businesses reduce customer attrition by understanding the patterns and reasons behind customers leaving. This project includes defining analysis objectives, collecting customer data, designing relevant visualizations in IBM Cognos, and building a predictive model.

ANALYSING OBJECTIVES

The specific objectives of predicting customer churn typically include:

1. Identifying Potential Churners:

The primary goal is to accurately identify customers who are at risk of churning, i.e., those likely to stop using a product or service. This involves creating a predictive model that assigns churn probability scores to individual customers.

2. Minimizing Churn Rate:

The aim is to reduce the overall churn rate by taking proactive measures to retain high-risk customers. This might involve targeted marketing campaigns, personalized offers, or improved customer service.

3. Understanding Key Factors:

To address churn effectively, it's crucial to identify the key drivers or factors contributing to churn. This requires analysing customer data to uncover patterns and insights, such as customer demographics, usage behaviour, and customer interactions.

4. Segmenting Customer Base:

Creating customer segments based on churn likelihood and behaviour can help tailor retention strategies. For example, different approaches may be needed for long-term loyal customers compared to those who exhibit sudden changes in behaviour.

5. Evaluating Intervention Strategies:

Assessing the effectiveness of churn prevention strategies is essential. Tracking changes in churn rates after implementing interventions can help refine and optimize retention efforts.

6. Predicting Future Revenue Loss:

By quantifying the potential revenue loss associated with churn, businesses can better understand the financial impact and allocate resources accordingly.

Overall, predicting customer churn aims to proactively manage customer retention by leveraging data-driven insights and strategies to mitigate churn and maximize customer lifetime value.

DATA COLLECTION

The dataset contains 7043 rows and 21 features. Each row represents a customer, each column contains the customer's attributes described in the column Metadata. The data set includes information about:

- Customers who left within the last month — the column is called Churn — this is the target Feature.
- Services that each customer has signed up for — phone, multiple lines, internet, online security, online backup, device protection, tech support, and streaming TV and movies
- Customer account information — how long they've been a customer, contract, payment method, paperless billing, monthly charges, and total charges.
- Demographic info about customers — gender, age range, and if they have partners and Dependent.

PREDICTIVE MODELING

Predictive modeling for customer churn prediction can benefit from both ensemble models and feature engineering.

ENSEMBLE MODELS

1. Random Forest:

Start with Random Forest as a baseline ensemble model. It's robust and can handle various types of data effectively.

2. Gradient Boosting (e.g., XGBoost, LightGBM):

Implement gradient boosting algorithms. They often outperform other models in churn prediction tasks due to their ability to capture complex patterns in the data.

3. Voting Classifier:

Create an ensemble of multiple models (e.g., Random Forest, XGBoost, and Logistic

Regression) using a Voting Classifier. This can help combine the strengths of different algorithms.

4. Bagging and Boosting:

Experiment with bagging (e.g., Bagged Decision Trees) and boosting (e.g., AdaBoost) techniques to improve model performance further.

5. Stacking:

Consider stacking multiple models by training a meta-learner on the predictions of base models. This can capture higher-level patterns in the data.

FEATURE ENGINEERING

1. Feature Selection:

Use techniques like feature importance scores from ensemble models (e.g., Random Forest's feature importance) to identify the most relevant features.

2. Interaction Features:

Create interaction features by combining existing features. For example, can be multiply usage frequency by customer tenure to capture the long-term value of customers.

3. Time-Based Features:

Incorporate time-based features to capture temporal patterns. These could include the month of subscription, the day of the week, or seasonality effects.

4. Customer Segmentation:

Segment customer base based on behavior, demographics, or other factors and use segment-specific features or models.

5. Feature Scaling and Normalization:

Ensure that numerical features are scaled or normalized to prevent models from being biased toward features with larger magnitudes.

6. Text Data:

If we have text data (e.g., customer feedback), perform text preprocessing and use techniques like TF-IDF or word embeddings to extract valuable information.

7. Handling Imbalanced Data:

If dataset is imbalanced (churned customers are much fewer than retained ones), consider techniques like oversampling, undersampling, or synthetic data generation to balance it.

8. Feature Importance Analysis:

Continuously analyze feature importance and their impact on model performance. Remove irrelevant or redundant features.

9. Feature Engineering Feedback Loop:

Iteratively refine feature engineering process based on model performance. Test different feature combinations and transformations.

The data has to be split into training, validation, and test sets for model evaluation. Additionally, have to perform hyperparameter tuning for ensemble models to optimize their performance.

In practice, a combination of ensemble models and feature engineering often yields the best results for customer churn prediction. It allows us to harness the predictive power of sophisticated algorithms while fine-tuning the feature set to capture the most meaningful signals in the data.

ADVANCED REGRESSION TECHNIQUES

When dealing with advanced regression techniques like ensemble models and feature engineering for a customer churn prediction project, you can significantly enhance the predictive accuracy and robustness of your model. Here's how to incorporate these techniques:

1.Ensemble Models

a. Random Forest Ensemble

Implement a Random Forest Classifier to leverage the power of multiple decision trees. It's robust against overfitting and captures complex relationships.

b. Gradient Boosting Ensemble(e.g., XGBoost, LightGBM, CatBoost)

These algorithms create an ensemble of decision trees and optimize them sequentially. They often provide state-of-the-art performance in classification tasks like churn prediction.

c. Stacking Ensemble

Combine predictions from multiple models (e.g., logistic regression, random forest, gradient boosting) to create a meta-model. This meta-model can offer improved predictive performance.

2. Support Vector Machine

A Support Vector Machine is a supervised machine learning algorithm used for classification and regression tasks. It works by finding the optimal hyperplane that best separates data points into different classes. The key concept is SVM is to maximize the margin which is the distance between the hyperplane and the nearest data points.

3. Nearest Neighbour Classifier

A Nearest Neighbor Classifier also known as the k-Nearest Neighbor classifier, is a simple and intuitive machine learning algorithm used for classification and regression tasks. It operates based on the principle of proximity: objects that are close to each other in a feature space are likely to belong to the same class.

4. Evaluation Metrics

Choose appropriate evaluation metrics for your churn prediction task, such as accuracy, precision, recall, F1-score, ROC AUC, or customer lifetime value (CLV) metrics, depending on your business objectives.

5. Model Interpretability

Consider techniques like SHAP values or Partial Dependency Plots to interpret and explain the predictions of your ensemble models, especially if interpretability is crucial for decision-making.

6. Regularization Techniques

Apply regularization methods like L1 (Lasso) or L2 (Ridge) regularization to prevent overfitting in your models.

By combining ensemble techniques with effective feature engineering and thorough model evaluation, we can create a robust and accurate customer churn prediction model that will help your business proactively retain valuable customers.

MODEL EVALUATION AND SELECTION

Selecting and evaluating a model for a customer churn prediction project involves several steps. Here's a high-level overview:

1. Data Preparation:

- Collect and clean your data, handling missing values and outliers.
- Feature engineering: Create relevant features that might improve model performance.
- Split your data into training, validation, and test sets.

2. Selecting Algorithms:

- Choose a set of algorithms to evaluate.
- Common choices include logistic regression, decision trees, random forests, support vector machines, and gradient boosting methods like XGBoost or LightGBM.

3. Evaluation Metrics:

- Define appropriate evaluation metrics for your problem.
- For churn prediction, commonly used metrics include accuracy, precision, recall, F1-score, ROC AUC, and lift.

4. Baseline Model:

- Start with a simple baseline model.
- It can be a logistic regression model or a decision tree, which provides a benchmark for more complex models.

5. Model Training:

- Train each selected model using the training dataset.
- Tune hyperparameters using techniques like grid search or random search.

6. Model Evaluation:

- Evaluate each model's performance on the validation dataset using the chosen metrics.
- Consider model interpretability for business insights.

MODEL TRAINING

Model training is the process of teaching a machine learning model to predict customer churn. It involves in the field of marketing, scoring is a technique that enables one to assign a score to a customer or a prospect. The corresponding score generally reflects the probability that a customer or prospect will answer a marketing solicitation. Predictive scoring can be used to identify customers at high risk of attribution.

Choose A Machine Learning Algorithm

There are a number of different machine learning algorithms that can be used for customer churn prediction such as Logistic Regression, Bayes Algorithm, Decision Trees, Support Vector Machines, Linear Discriminant Analysis.

MODEL INTERPRETABILITY

The demand for interpretability increases when there is a large cost for incorrect predictions, especially in high-risk applications. Based on the model complexity, methods for model interpretability can be classified into intrinsic analysis and post hoc analysis. The model itself becomes the source of knowledge instead of the data. Interpretability makes it possible to extract this additional knowledge captured by the model. Machine learning models take on real-world tasks that require safety measures and testing. It can help improve interpretability by reducing the model's complexity and avoiding unnecessary features. You can use regularization techniques such as Lasso, Ridge, or Elastic Net to shrink or eliminate the coefficients of less important features, making the model more sparse and simple.

DEPLOYMENT AND PREDICTION

Here we will train an optimized (treebased) model which will predict if a customer will or won't churn.

Data Preparation

Before we start training a model we must prepare our data. Different steps that we can undertake:

- Encode all categorical data (watch out with one hot encoding and tree-based models...).
- Scale data
- Check correlation matrix to extract the most influential features.
- Generate new columns from data.
- Upsample the imbalanced dataset (SMOTE/ADASYN).

In this notebook we shall focus on the upsampling method. The data wrangling performed is to make sure that the upsampling is performed in a correct manner.

SMOTE (Synthetic Minority Oversampling Technique)

We saw that our dataset was imbalanced. This could gives problems when creating a classification model since it might not learn the decision boundary. This ofcourse an be solved with upsampling.

One technique used for this is SMOTE, this technique creates new synthetic samples which can be used for training.

SMOTE first selects a minority class instance a at random and finds its k nearest minority class neighbors. The synthetic instance is then created by choosing one of the k nearest neighbors b at random and connecting a and b to form a line segment in the feature space. The synthetic instances are generated as a convex combination of the two chosen instances a and b .

PROGRAM

Customer Churn Prediction

Importing Dependencies

In[1] :

```
From mpl_toolkits.mplot3d import Axes3D
```

```
From sklearn.preprocessing import StandardScaler
```

```
Import matplotlib.pyplot as plt
```

```
Import numpy as np
```

```
Import os
```

```
Import pandas as pd
```

```
Print(os.listdir(['WA_Fn-UseC_-Telco-Customer-Churn.csv']))
```

In[2]:

```
path = '../input/telco-customer-churn/WA_Fn-UseC_-Telco-Customer-Churn.csv'
```

```
data = pd.read_csv(path)
```

```
data.head()
```

Out[2]:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSec
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No

Model 1 - Logistic Regression

In[3]:

```
Def logistic_regression(frame, grid):  
  
    Logit = Model(LogisticRegression, frame, metrics)  
  
    Print_result(logit.grid_search('roc_auc', grid), "cross-validation")  
  
    Print_result(logit.train_and_evaluate_test(), "test set")  
  
    Print("Best hyperparameters:", logit.hyperparameters)  
  
Logistic_regression(df_processed, {'C': np.logspace(-4, 4, 100, base=10)})
```

Out[3]:

```
{'random_state': 50}  
  
| roc_auc: 0.8463 | accuracy: 0.8046 – cross-validation  
  
| roc_auc: 0.8399 | accuracy: 0.7999 – test set  
  
Best hyperparameters: {'C': 1.3219411484660315}
```

In[4]:

```
Logit_grid = {'C': np.linspace(0.02, 3, 150)}  
  
Logistic_regression(df_processed, logit_grid)
```

Out[4]:

```
{'random_state': 50}  
  
| roc_auc: 0.8463 | accuracy: 0.8046 – cross-validation  
  
| roc_auc: 0.8399 | accuracy: 0.7999 – test set  
  
Best hyperparameters: {'C': 1.3800000000000001}
```

Model 2-Support Vector Machine

Rbf Kernel

In[5]:

```
Def svc_rbf(frame, grid):
```

```
    Rbf = Model(SVC, frame, metrics, fixed_params= {'kernel': 'rbf', 'probability': True})
```

```
    Print_result(rbf.grid_search('roc_auc', grid, num_folds=4), "cross-validation")
```

```
    Print_result(rbf.train_and_evaluate_test(), "test set")
```

```
    Print("Best hyperparameters:", rbf.hyperparameters)
```

```
Grid_rbf = {'C': np.logspace(-4, 1, 10, base=2), 'gamma': np.logspace(-6, 2, 10, base=2)}
```

```
Svc_rbf(df_processed, grid_rbf)
```

Out[5]:

```
{'kernel': 'rbf', 'probability': True, 'random_state': 50}
```

```
| roc_auc: 0.8365 | accuracy: 0.7969 – cross-validation
```

```
| roc_auc: 0.8352 | accuracy: 0.7999 – test set
```

```
Best hyperparameters: {'C': 0.13500746736153826, 'gamma': 0.015625}
```

Model 3 - Nearest Neighbors Classifier

In[6]:

```
Def knn_clf(frame, grid):
```

```
    Knn = Model(KNeighborsClassifier, frame, metrics)
```

```
    Print_result(knn.grid_search('roc_auc', grid), "cross-validation")
```

```
    Print_result(knn.train_and_evaluate_test(), "test set")
```

```
    Print("Best hyperparameters:", knn.hyperparameters)
```

```
Knn_clf(df_processed, {'n_neighbors': [i for i in range(10, 50, 2)]})
```

Out[6]:

```
{}
```

| roc_auc: 0.8360 | accuracy: 0.7984 – cross-validation

| roc_auc: 0.8324 | accuracy: 0.7906 – test set

Best hyperparameters: {'n_neighbors': 48}

Duplicate Values

In[7]:

```
data.duplicated().sum()
```

Out[7]:

```
0
```

In[8]:

```
data.drop('customerID', axis=1, inplace=True)
```

```
data.head()
```

Out[8]:

	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSup
0	Female	0	Yes	No	1	No	No phone service	DSL	No	Yes	No	No
1	Male	0	No	No	34	Yes	No	DSL	Yes	No	Yes	No
2	Male	0	No	No	2	Yes	No	DSL	Yes	Yes	No	No
3	Male	0	No	No	45	No	No phone service	DSL	Yes	No	Yes	Yes
4	Female	0	No	No	2	Yes	No	Fiber optic	No	No	No	No

Renaming Column Names

In[9]:

```
column_list = list(data.columns)
```

```

for i, col in enumerate(column_list):

    column_list[i] = inflection.underscore(column_list[i]).replace(' ', '_')

data.columns = column_list

data.head()

```

Out[9]:

	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSup
0	Female	0	Yes	No	1	No	No phone service	DSL	No	Yes	No	No
1	Male	0	No	No	34	Yes	No	DSL	Yes	No	Yes	No
2	Male	0	No	No	2	Yes	No	DSL	Yes	Yes	No	No
3	Male	0	No	No	45	No	No phone service	DSL	Yes	No	Yes	Yes
4	Female	0	No	No	2	Yes	No	Fiber optic	No	No	No	No

Data Types

General Data Type

In[10]:

```

# numerical

column_numerical = ['tenure', 'monthly_charges', 'total_charges']

# categorical

column_categorical = list(data.columns)

column_categorical.remove('tenure')

column_categorical.remove('monthly_charges')

column_categorical.remove('total_charges')

data_type_general = dict()

for col in data.columns:

    if col in column_numerical:

        data_type_general[col] = 'numerical'

```

```
else:
```

```
    data_type_general[col] = 'categorical'
```

```
tmp = pd.Series(data_type_general)
```

```
data_type_general = pd.DataFrame(tmp).T.rename({0:'general data types'})
```

```
data_type_general
```

Out[10]:

	gender	senior_citizen	partner	dependents	tenure	phone_service	multiple_lines	internet_service	online_security	online_backup	device_p
general data types	categorical	categorical	categorical	categorical	numerical	categorical	categorical	categorical	categorical	categorical	categoric

Python Data Type

In[11]:

```
data_type_python = pd.DataFrame(data.dtypes).T.rename({0:'python data types'})
```

```
data_type_python
```

out[11]:

	gender	senior_citizen	partner	dependents	tenure	phone_service	multiple_lines	internet_service	online_security	online_backup	device_protection
python data types	object	int64	object	object	int64	object	object	object	object	object	object

In[12]:

```
data['total_charges'].isna().sum()
```

Out[12]:

```
0
```

In[13]:

```
data.loc[488, 'total_charges']
```

Out[13]

```
''
```

In[14]:

```
data.loc[data['tenure'] == 0, ['tenure', 'total_charges', 'churn']]
```

Out[14]:

	tenure	total_charges	churn
488	0		No
753	0		No
936	0		No
1082	0		No
1340	0		No
3331	0		No
3826	0		No
4380	0		No
5218	0		No
6670	0		No
6754	0		No

In[15]:

```
data['total_charges']=pd.to_numeric(data['total_charges'], errors='coerce')
```

```
data['total_charges'].isna().sum()
```

Out[15]:

11

In[16]:

```
data['total_charges'].fillna(0, inplace=True)
```

```
data['total_charges'].isna().sum()
```

Out[16]:

0

In[17]:

```
data['total_charges'] = data['total_charges'].astype(float)
```

```
data.dtypes
```

Out[17]:

```
gender           object
senior_citizen   int64
partner          object
dependents       object
tenure           int64
phone_service    object
multiple_lines   object
internet_service object
online_security  object
online_backup    object
device_protection object
tech_support     object
streaming_tv     object
streaming_movies object
contract         object
paperless_billing object
payment_method   object
monthly_charges  float64
total_charges    float64
churn            object
dtype: object
```

In[18]:

```
data_type_python=pd.DataFrame(data.dtypes).T.rename({0:'python
data types'})
```

```
data_type_python
```

Out[18]:

	gender	senior_citizen	partner	dependents	tenure	phone_service	multiple_lines	internet_service	online_security	online_backup	device_protection
python data types	object	int64	object	object	int64	object	object	object	object	object	object

Total Data

In[19]:

```
data_count = pd.DataFrame(data.count()).T.rename({0:'total data'})
```

data_count

Out[19]:

[illegible]

In[20]:

```
Print("Total number of records in training dataset:", train_data.shape)
```

Out[20]:

Total number of records in training dataset: (7043, 21)

In[21]:

```
data_null_total=pd.DataFrame(data.isna().sum()).T.rename({0:'total null'})
```

data_null_total

Out[21]:[illegible]

Null Values Percentage

In[22]:

```
data_null_percentage=pd.DataFrame(100*data.isna().sum()/data.shape[0]).T.rename({0:'percentage null'})
```

data_null_percentage

Out[22]:

	gender	senior_citizen	partner	dependents	tenure	phone_service	multiple_lines	internet_service	online_security	online_backup	device_protection
percentage null	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Data Variation

Numerical Data

In[23]:

```
variation_numerical = dict()
```

```
for col in column_numerical:
```

```
    tmp = f'{data[col].min()} - {data[col].max()}'
```

```
    variation_numerical[col] = tmp
```

```
tmp = pd.Series(variation_numerical)
```

```
data_variation_numerical=pd.DataFrame(tmp).T.rename({0:'data variation'})
```

data_variation_numerical

Out[23]:

	tenure	monthly_charges	total_charges
data variation	0 - 72	18.25 - 118.75	0.0 - 8684.8

Categorical Data

In[24]:

```
variation_categorical = dict()

for col in column_categorical:

    tmp = data[col].unique().tolist()

    tmp.sort()

    variation_categorical[col] = ', '.join(str(item) for item in tmp)

tmp = pd.Series(variation_categorical)

data_variation_categorical=pd.DataFrame(tmp).T.rename({0:'data variation'})

data_variation_categorical
```

Out[24]:

	gender	senior_citizen	partner	dependents	phone_service	multiple_lines	internet_service	online_security	online_backup	device_protection	tech_si
data variation	Female, Male	0, 1	No, Yes	No, Yes	No, Yes	No, No phone service, Yes	DSL, Fiber optic, No	No, No internet service, Yes	No, No internet service, Yes	No, No internet service, Yes	No, No internet service, Yes

Overall Data Variation

In[25]:

```
data_variation=pd.concat([data_variation_numerical,data_variation_categorical], axis=1)

data_variation
```

Out[25]:

	tenure	monthly_charges	total_charges	gender	senior_citizen	partner	dependents	phone_service	multiple_lines	internet_service	online_security
data variation	0 - 72	18.25 - 118.75	0.0 - 8684.8	Female, Male	0, 1	No, Yes	No, Yes	No, Yes	No, No phone service, Yes	DSL, Fiber optic, No	No, No internet service, Yes

In[26]:

```
pd.concat([data_variable_numerical.rename({'data variation': 'range'}), data_variable_
categorical.rename({'data variation': 'variation'})], axis=0).fillna('').reindex(data.columns,
axis=1).T
```

Out[26]:

	range	variation
gender	-	Female, Male
senior_citizen	-	0, 1
partner	-	No, Yes
dependents	-	No, Yes
tenure	0 - 72	-
phone_service	-	No, Yes
multiple_lines	-	No, No phone service, Yes
internet_service	-	DSL, Fiber optic, No
online_security	-	No, No internet service, Yes
online_backup	-	No, No internet service, Yes
device_protection	-	No, No internet service, Yes
tech_support	-	No, No internet service, Yes
streaming_tv	-	No, No internet service, Yes
streaming_movies	-	No, No internet service, Yes
contract	-	Month-to-month, One year, Two year
paperless_billing	-	No, Yes
payment_method	-	Bank transfer (automatic), Credit card (automatic), Electronic check, Mailed check
monthly_charges	18.25 - 118.75	-
total_charges	0.0 - 8684.8	-
churn	-	No, Yes

Redundant Values

In[27]:

```
data.replace({'No phone service':'No', 'No internet service':'No'}, inplace=True)
```

```
data['senior_citizen'].replace({0:'No', 1:'Yes'}, inplace=True)
```

In[28]:

```
variation_categorical = dict()

for col in column_categorical:

    tmp = data[col].unique().tolist()

    tmp.sort()

    variation_categorical[col] = ', '.join(str(item) for item in tmp)

tmp = pd.Series(variation_categorical)

data_variation_categorical=pd.DataFrame(tmp).T.rename({0:'data variation'})

data_variation=pd.concat([data_variation_numerical,data_variation_categorical], axis=1)

data_variation
```

Out[28]:

	tenure	monthly_charges	total_charges	gender	senior_citizen	partner	dependents	phone_service	multiple_lines	internet_service	online_security
data variation	0 - 72	18.25 - 118.75	0.0 - 8684.8	Female, Male	No, Yes	No, Yes	No, Yes	No, Yes	No, Yes	DSL, Fiber optic, No	No, Yes

Summary of Dataset Info

In[29]:

```
data_info = pd.concat([data_type_general, data_type_python,

                        data_count, data_null_total,

                        data_null_percentage.round(2), data_variation],

                        axis=0)

data_info = data_info.reindex(data.columns, axis=1)

data_info
```

Out[29]:

	gender	senior_citizen	partner	dependents	tenure	phone_service	multiple_lines	internet_service	online_security	online_backup	device
general data types	categorical	categorical	categorical	categorical	numerical	categorical	categorical	categorical	categorical	categorical	categorical
python data types	object	int64	object	object	int64	object	object	object	object	object	object
total data	7043	7043	7043	7043	7043	7043	7043	7043	7043	7043	7043
total null	0	0	0	0	0	0	0	0	0	0	0
percentage null	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
data variation	Female, Male	No, Yes	No, Yes	No, Yes	0 - 72	No, Yes	No, Yes	DSL, Fiber optic, No	No, Yes	No, Yes	No, Yes

EXPLORATORY DATA ANALYSIS

Demographic Info About Customers

1- Gender

In[30]:

```
gndr_crn=df.groupby(['gender','Churn']).apply(lambda x:x['Churn'].count()).reset_index  
                                                (name='Counts')
```

In[31]:

linkcode

gndr_crn

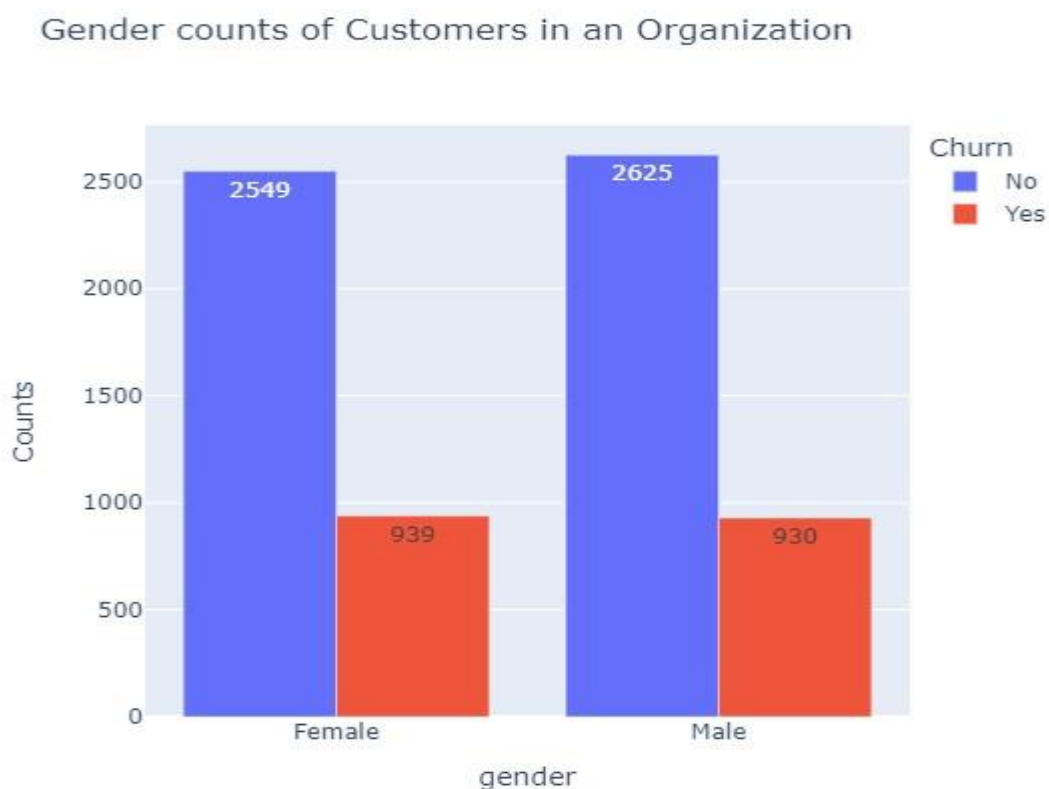
Out[31]:

	gender	Churn	Counts
0	Female	No	2549
1	Female	Yes	939
2	Male	No	2625
3	Male	Yes	930

In[32]:

```
fig=px.bar(gndr_crn,x='gender',y='Counts',color='Churn',text_auto=True,  
           title='Gender counts of Customers in an Organization')  
  
fig.update_layout(barmode='group', xaxis_tickangle=0)  
  
fig.show()
```

Out[32]:



2- age

In[33]:

```
snrctzn_crn=df.groupby(['SeniorCitizen','Churn']).apply(lambdax:x['Churn'].count()).reset_in  
dex(name='Counts')
```

```
fig=px.bar(snretzn_crn,x='SeniorCitizen',y='Counts',color='Churn',
text_auto=True,title='Gender counts of Customers in an Organization')

fig.update_layout(barmode='group', xaxis_tickangle=0)

fig.show()
```

Out[33]:



In[34]:

```
per=np.array([(4508/(4508+1393)),(1393/(4508+1393)),(666/(666+476)),(476/(666+476))]).
round(2)
```

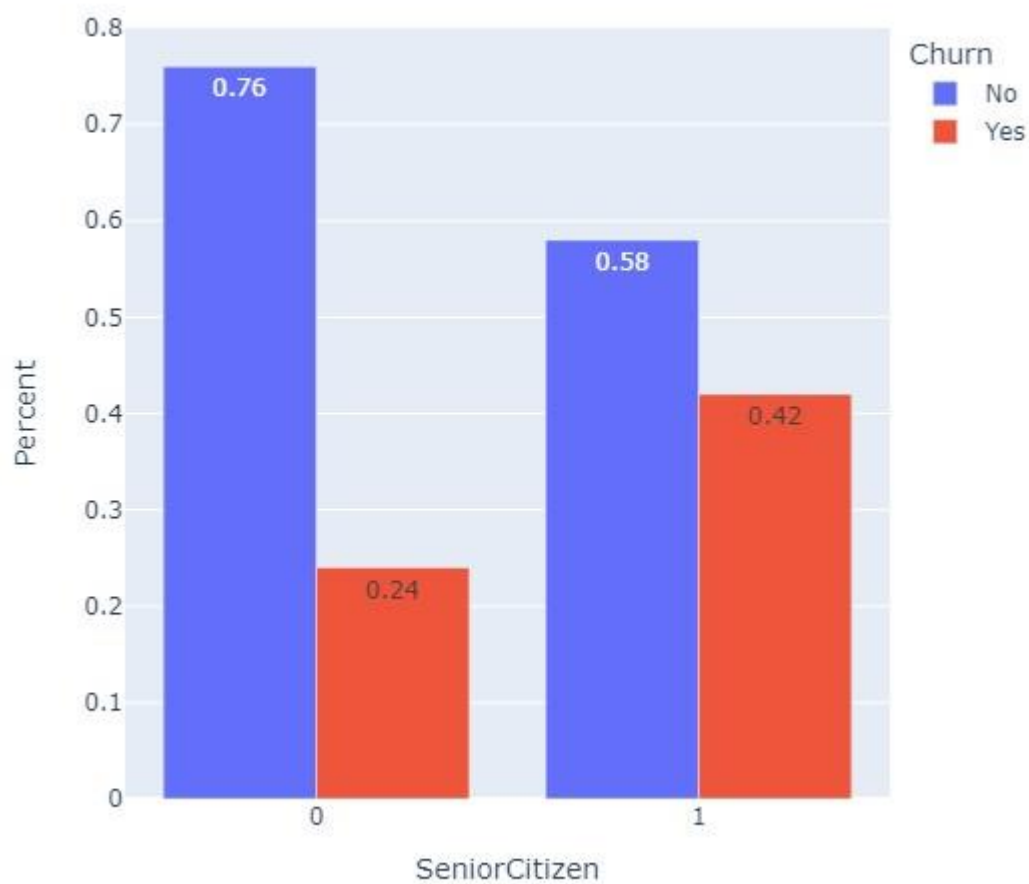
```
snrctzn_crn['Percent']=per
```

```
fig=px.bar(snrctzn_crn,x='SeniorCitizen',y='Percent',color='Churn', text_auto=True)
```

```
fig.update_layout(barmode='group', xaxis_tickangle=0)
```

```
fig.show()
```

Out[34]:

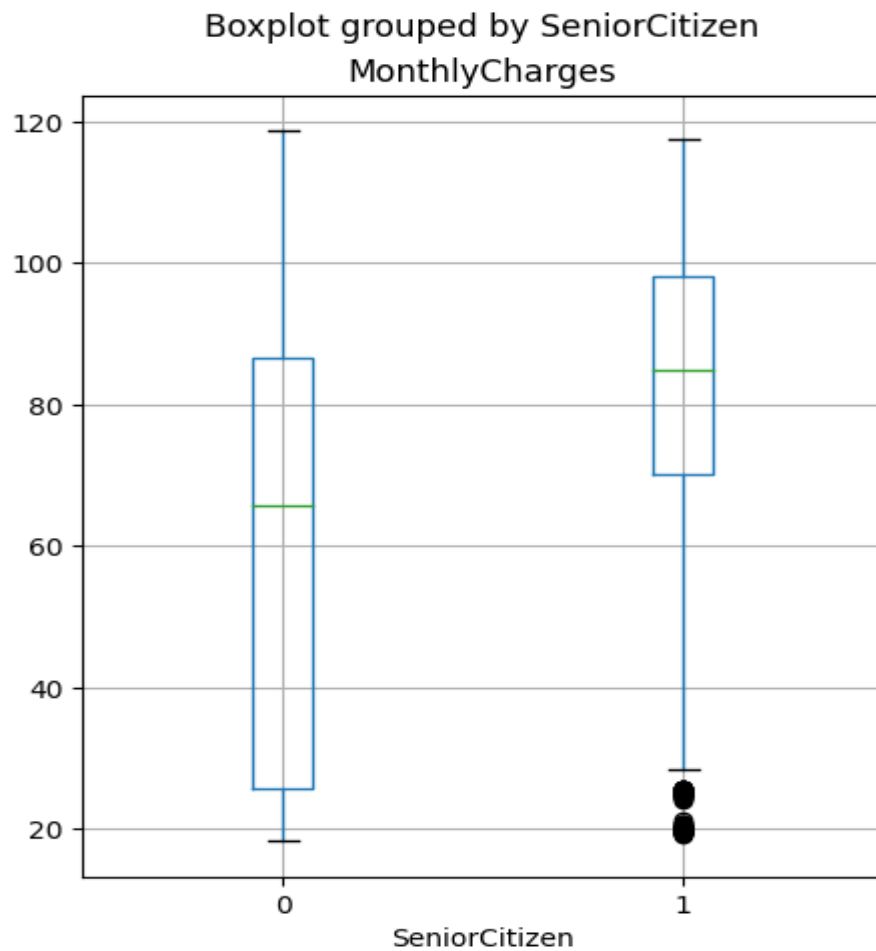


In[35]:

```
df.boxplot(column='MonthlyCharges', by='SeniorCitizen', figsize=(5,6))
```

Out[35]:

```
<Axes: title={'center': 'MonthlyCharges'}, xlabel='SeniorCitizen'>
```



3- partner

In[36]:

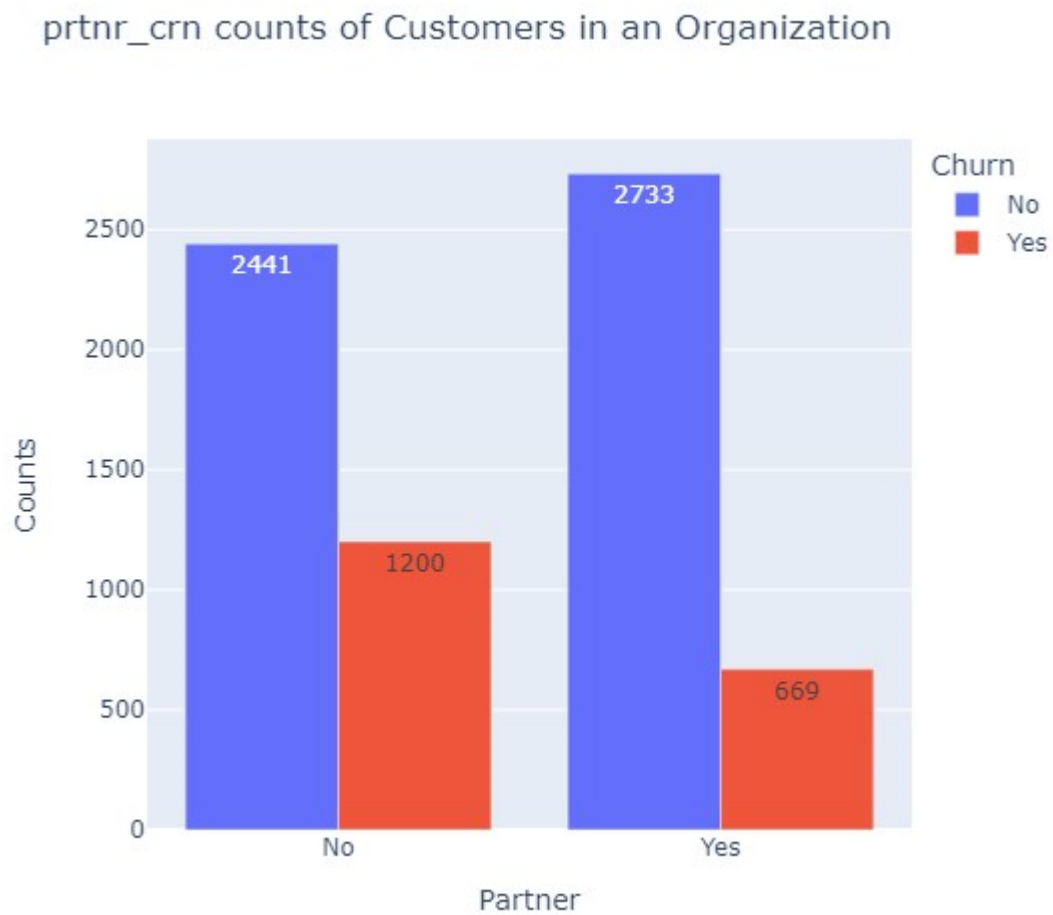
```
prtnr_crn=df.groupby(['Partner','Churn']).apply(lambdax:x['Churn'].count()).reset_index(
                                                    name ='Counts')

fig=px.bar(prtnr_crn,x='Partner',y='Counts',color='Churn',text_auto=True,title='prtnr_crn
                                                    counts of Customers in an Organization')

fig.update_layout(barmode='group', xaxis_tickangle=0)

fig.show()
```

Out[36]:



In[37]:

```
per=np.array([(2441/(2441+1200)),(1200/(2441+1200)),(2733/(669+2733)),(669/(669+2733))]).round(2)
```

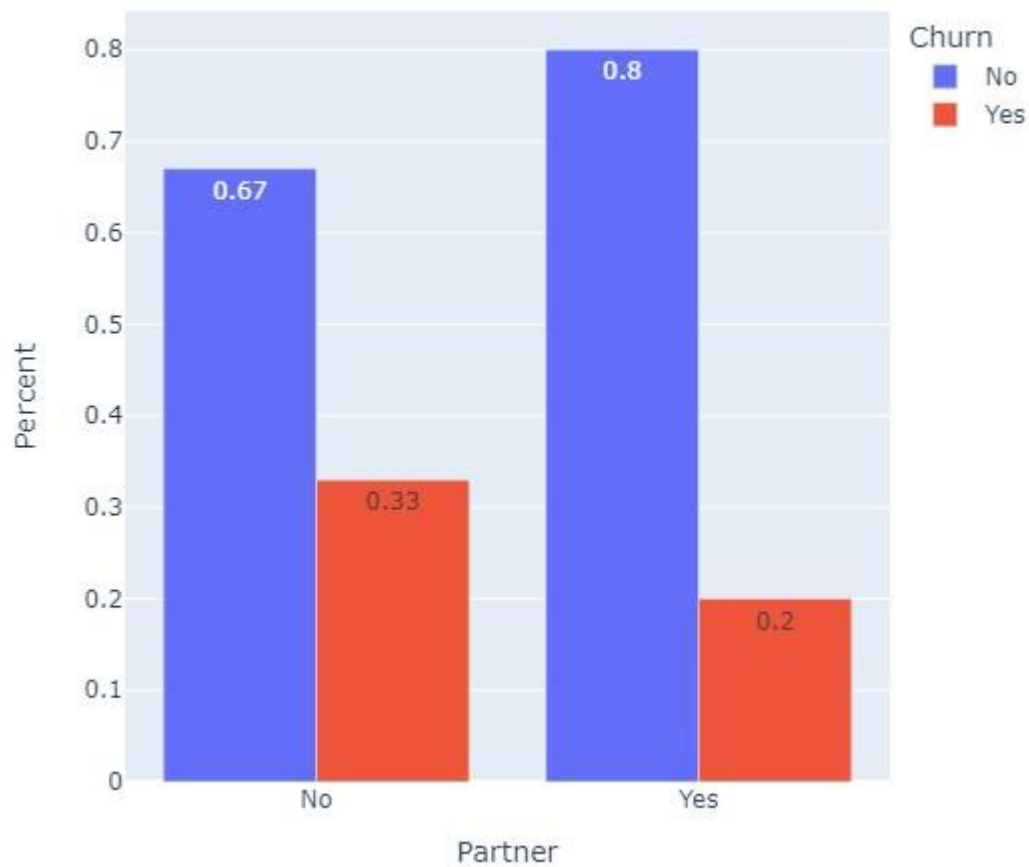
```
prtnr_crn['Percent']=per
```

```
fig=px.bar(prtnr_crn,x='Partner',y='Percent',color='Churn', text_auto=True)
```

```
fig.update_layout(barmode='group', xaxis_tickangle=0)
```

```
fig.show()
```

Out[37]:

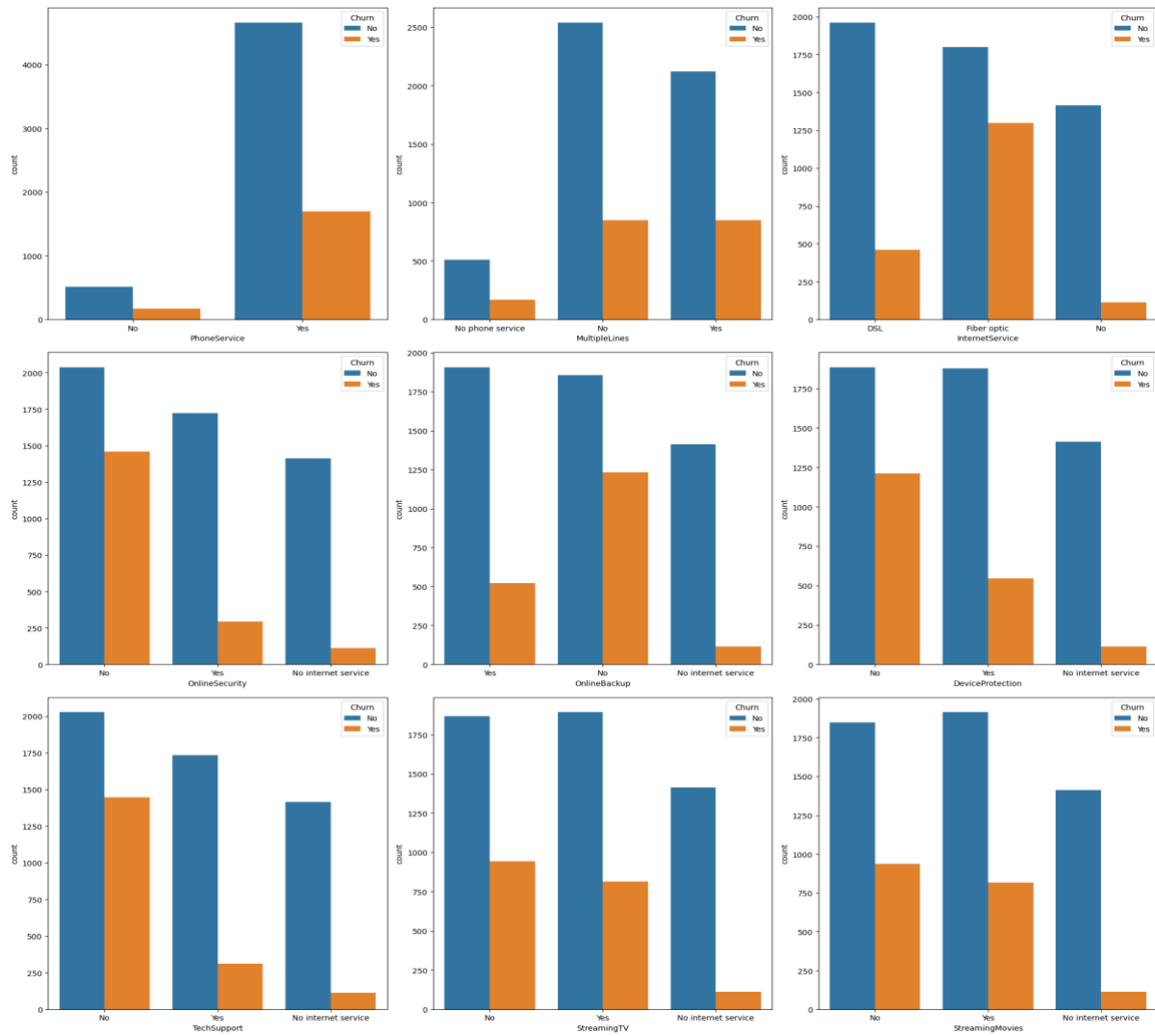


In[38]:

```
col=['PhoneService','MultipleLines','InternetService','OnlineSecurity',  
     'OnlineBackup','DeviceProtection','TechSupport','StreamingTV','StreamingMovies']  
  
fig,axs=plt.subplots(3,3,figsize=(20,20))  
  
axs=axs.flatten()  
  
for i,var in enumerate(col):  
    sns.countplot(data=df,x=var,hue='Churn',ax=axs[i])  
  
plt.tight_layout()
```

plt.show()

Out[38]:



CONCLUSION

In conclusion, the customer churn prediction project has provided valuable insights and outcomes that can significantly benefit our business. By leveraging advanced machine learning techniques, we have developed a robust model that accurately identifies potential churners. This predictive capability allows us to proactively engage with at-risk customers, implement targeted retention strategies, and ultimately reduce customer churn. The successful implementation of this project demonstrates the power of data-driven decision-making in retaining our valuable customers and ensuring the long-term success of our business.

FUTURE ENHANCEMENT

The future of churn prediction lies in advancements in machine learning algorithms, enabling more accurate predictions and a deeper understanding of customer behavior. Integration of artificial intelligence (AI) in churn prediction will enhance the automation of data analysis and decision-making processes. Predictive analytics will continue to evolve, incorporating real-time churn analysis, enabling businesses to take immediate action and proactively retain customers.